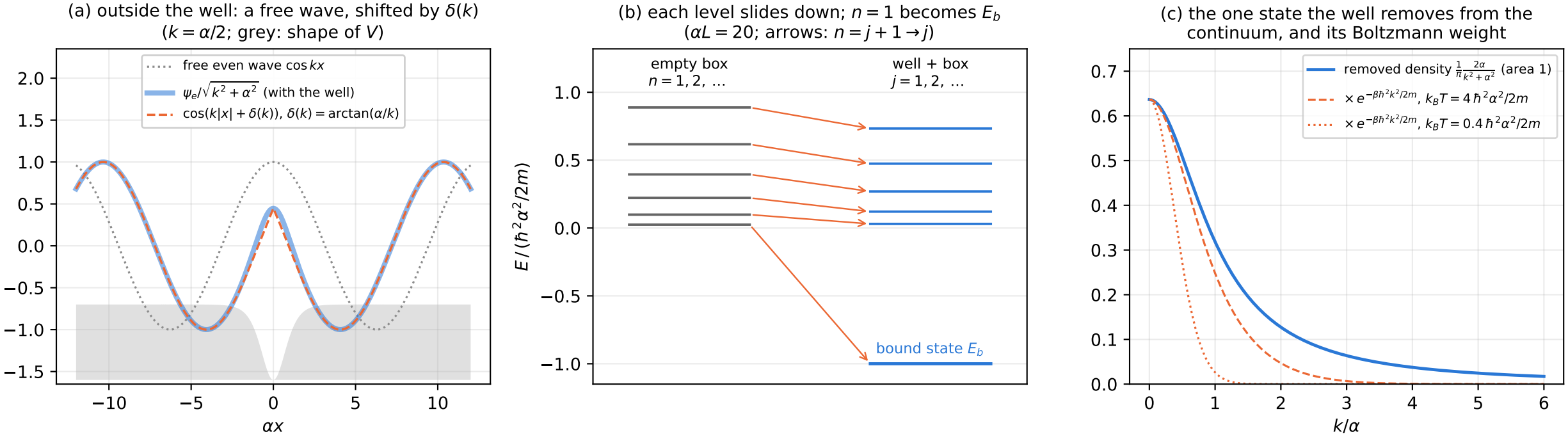

(a) outside the well: a free wave, shifted by $\delta(k)$
($k = \alpha/2$; grey: shape of $V$)
free even wave $\cos kx$
$\psi_e/\sqrt{k^2+\alpha^2}$ (with the well)
$\cos(k|x| + \delta(k))$, $\delta(k) = \arctan(\alpha/k)$
$\alpha x$
(b) each level slides down; $n = 1$ becomes $E_b$
($\alpha L = 20$; arrows: $n = j + 1 \to j$)
empty box
$n = 1, 2, \ldots$
well + box
$j = 1, 2, \ldots$
bound state $E_b$
$E\,/\,(\hbar^2\alpha^2/2m)$
(c) the one state the well removes from the
continuum, and its Boltzmann weight
removed density $\frac{1}{\pi}\frac{2\alpha}{k^2+\alpha^2}$ (area 1)
$\times\, e^{-\beta\hbar^2k^2/2m}$, $k_BT = 4\,\hbar^2\alpha^2/2m$
$\times\, e^{-\beta\hbar^2k^2/2m}$, $k_BT = 0.4\,\hbar^2\alpha^2/2m$
$k/\alpha$